(8)

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

I Dinesh Bharadwaj (192572028) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Bh

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	1	1	1	1
Organization	15%				
Accuracy	10%				
Clarity	5%				

Total Marks 20 **/40**

Signature of the Faculty

FTP Application:-

Data = 3GB, Throughput = 120 Mbps, Request size = 3MB

Convert data:- 3GB = 3000MB

No.of Requests:- 3000 ÷ 3 = 1000

Transmission Time 3000 * 8 = ~~1000~~ 24000 Mb

24000 ÷ 120 = 200 Seconds

The Application layer provides uses ~~R~~ services such as authentication, file transfer, error reparting, and resource sharing ensuring reliable communication.

End to End delay:-

Application = 4ms

Transpart = 3ms Data link = 2ms Propagation = 18ms network = 5ms physical = 1ms

Transmission = 12ms

Total delay = 4 + 3 + 5 + 2 + 1 + 18 + 12

= 45ms

9. Network efficiency:-

data = 80MB

Frame size = 1600 bytes

overhead = 80 byte/frame

convert data · 80,000,000 bytes

no. of frames :- 1600 − 80 = 1520

$$\frac{80\,000\,000}{1520} = 52,632$$

52,632 × 80 = 4210560

= 4.21 MB

Transmission efficiency :- $\frac{1520}{1000} \times 100 = 95\%$

10. Healthcare IOT communication

original data = 240MB   segment size = 1400

compression = 30%.   Frame overhead = 60

Step 1:- convert   240 × 0.70 = 168 MB

Step 2:- convert 168000 000 bytes

Step 3:- segments $\frac{168000000}{1400} = 120000$

Protocol overhead :- 120000 × 60 = 7200000 bytes

Total physical layer data   168 + 7.2 = 175.2 MB

Presentation layer compresses and encrypts medical records for efficient and secure transmission